

Use decimal notation in conversions

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Write 845 mm in metres. Copy or complete the first step.

2. Use 5 miles is about 8 km. Estimate 30 miles in kilometres.

3. A 12.5 m roll is cut into 40 cm pieces. How many complete pieces can be cut?

4. Miro says: Miro multiplies whenever converting, regardless of the direction between units. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Write 845 mm in metres. Copy or complete the first step.

Answer 0.845 m.

2. Use 5 miles is about 8 km. Estimate 30 miles in kilometres.

Answer About 48 km.

3. A 12.5 m roll is cut into 40 cm pieces. How many complete pieces can be cut?

Answer 31 complete pieces, with 10 cm left.

4. Miro says: Miro multiplies whenever converting, regardless of the direction between units. Is this correct? Explain.

Answer Use the size of the target unit to decide the direction, then check whether the numerical value should grow or shrink.

Use decimal notation in conversions

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Write 845 mm in metres.

6. Check this result using an inverse, estimate or known fact: A 12.5 m roll is cut into 40 cm pieces. How many complete pieces can be cut?

2. Use 5 miles is about 8 km. Estimate 30 miles in kilometres.

7. Prove or explain your reasoning: Convert 4.375 km to metres.

3. A 12.5 m roll is cut into 40 cm pieces. How many complete pieces can be cut?

8. Identify and correct this misconception: Miro multiplies whenever converting, regardless of the direction between units.

4. Solve this independently and show every stage: Write 845 mm in metres.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: Use 5 miles is about 8 km. Estimate 30 miles in kilometres.

10. Write and solve a similar question to: A 12.5 m roll is cut into 40 cm pieces. How many complete pieces can be cut?

Teacher answers and checking prompts

1. Write 845 mm in metres.
Answer 0.845 m.
2. Use 5 miles is about 8 km. Estimate 30 miles in kilometres.
Answer About 48 km.
3. A 12.5 m roll is cut into 40 cm pieces. How many complete pieces can be cut?
Answer 31 complete pieces, with 10 cm left.
4. Solve this independently and show every stage: Write 845 mm in metres.
Answer 0.845 m.
5. Use a second method or representation for: Use 5 miles is about 8 km. Estimate 30 miles in kilometres.
Answer Expected result: About 48 km. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: A 12.5 m roll is cut into 40 cm pieces. How many complete pieces can be cut?
Answer Expected result: 31 complete pieces, with 10 cm left. A valid check must be shown.
7. Prove or explain your reasoning: Convert 4.375 km to metres.
Answer 4,375 m. The explanation must show why the method works.
8. Identify and correct this misconception: Miro multiplies whenever converting, regardless of the direction between units.
Answer Use the size of the target unit to decide the direction, then check whether the numerical value should grow or shrink.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: A 12.5 m roll is cut into 40 cm pieces. How many complete pieces can be cut?
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Use decimal notation in conversions

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Prove or explain your reasoning: Convert 4.375 km to metres.

6. Create a new example that tests the same idea as: A 12.5 m roll is cut into 40 cm pieces. How many complete pieces can be cut?

2. Prove the answer to this question, rather than only calculating it: Write 845 mm in metres.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: Use 5 miles is about 8 km. Estimate 30 miles in kilometres.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 31 complete pieces, with 10 cm left.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro multiplies whenever converting, regardless of the direction between units.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: Convert 4.375 km to metres.
Answer 4,375 m. The explanation must show why the method works.
2. Prove the answer to this question, rather than only calculating it: Write 845 mm in metres.
Answer Expected result: 0.845 m. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: Use 5 miles is about 8 km. Estimate 30 miles in kilometres.
Answer Expected result: About 48 km. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 31 complete pieces, with 10 cm left.
Answer One valid reconstruction is: A 12.5 m roll is cut into 40 cm pieces. How many complete pieces can be cut? Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro multiplies whenever converting, regardless of the direction between units.
Answer Use the size of the target unit to decide the direction, then check whether the numerical value should grow or shrink.
6. Create a new example that tests the same idea as: A 12.5 m roll is cut into 40 cm pieces. How many complete pieces can be cut?
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.